LOI: "Measurement and Evaluation of State and Local Housing Supply Reforms"

Thomas Davidoff, Robert Helsley, and Tsur Somerville, Sauder School of Business, UB

June 19, 2026

1 Introduction and Contribution

Jurisdictions across Canada and the U.S. have recently allowed "missing middle" infill housing, such as accessory dwelling units and multiplexes, in residential zones formerly restricted to single-family use. Several studies have documented that infill housing is less prevalent in more expensive neighbourhoods.[1] This is puzzling in that the right to add density should be more valuable in neighbourhoods where prices are higher.

To resolve this seeming contradiction between the value added of new residential density and the geography of infill uptake, we develop a parsimonious model emphasizing that different implementations of missing middle zoning feature different mixes of "addition" (adding to allowable square footage) and "division" (subdividing a fixed amount of square footage into multiple units). Addition is more valuable where price per square foot of built space is greatest. Division is more valuable where the elasticity of price in square footage is less positive, that is, where buyers do not pay much more for larger units, so that splitting a house into smaller units sacrifices little per-unit value.

Infill housing will thus be less prevalent in pricier neighbourhoods when the upzoning features relatively little addition and more division if (wealthier) households with greater willingness to pay for space are concentrated in expensive price-per-square-foot neighbourhoods. Preliminary analysis of housing transactions and permitting data in upzoned neighbourhoods of Vancouver, Portland, and Minneapolis provide mixed evidence to date on these theoretical contributions. Division-only policies generally have take-up spatially negatively correlated with price per square foot. Mixed addition-and-division reforms have mixed uptake.

The contribution of a fuller study will be to document whether our model accurately captures an important reason why uptake of infill projects has (seemingly inefficiently) been centered in less valued locations in which displacement may be a more salient concern. One

[1] Marantz, Elmendorf, and Kim (2023) find that accessory dwelling unit (ADU) placement in California is widespread, but concentrated in convenient, low- and moderate-rent neighbourhoods, and not in more expensive neighbourhoods. Similar findings are in Brueckner and Thomaz (2024), who find that ADU permitting rates are lower in higher-priced neighbourhoods, even after controlling for income and other factors. Huang et al (2023) analyze the construction of duplexes and triplexes in Seattle following a 2019 upzoning reform and find that these housing types are more likely to be built in lower-priced neighbourhoods. Finally, Dong and Hansz (2019) examine the impact of Portland's Residential Infill Project (RIP) on housing construction patterns and find that the new zoning has led to more infill development in lower-priced areas.

inference that might follow from our results is that for infill zoning to succeed in inducing more affordable housing forms in high-value locations, more "addition" of density will be required.

2 Description of the policy being evaluated

Below, we describe the evolution of the relevant policies and their details in each of three cities that have been our focus to date for academic research purposes. We expect the breadth of the study area to grow for the proposed funded research.

Minneapolis, MN USA

Minneapolis was moved away from single-family zoning, with the City Council voting in December 2018 to allow more than one unit on all single-family lots. The changes came into effect in January 2020 following approval by the region's Metropolitan Council in 2019. The rezoning divides the city into three major regions, with additional distinct policies for transit corridors. These corridors were rezoned to allow 3–6 storey buildings, with 10–30 storeys permitted close to transit stations. Additionally, in May 2021, off-street parking minimums were eliminated. The key variation for our projects comes from the fact that the rezoning of Interior areas 1 and 2 were division only reforms, whereas Interior-3 was a mixed addition and division reform. Our model predicts likelier positive sorting in Interior-3, where there is more addition, versus others only division

Portland, OR USA

Broad upzoning in Portland was triggered by the Oregon legislature's passage of House Bill 2001 in July 2019. This bill effectively banned exclusive single-family zoning and required jurisdictions with over 25,000 residents, among other changes, to allow up to four units per lot. Cities were given until June 2022 to update their zoning codes to meet the bill's requirements. Portland's upzoning proceeded in two rounds, with the first round applying to zones nearer the city centre and the second round extending the changes to the remaining single-family zones. The first round of reforms (RIP-1) was effective in August 2021 and applied to zones R2.5, R5, and R7, which reflect allowed lot sizes of 2,500, 5,000, and 7,000 sq ft, respectively. Allowed FSR varied by zone and by the number of units built per lot, with an additional 0.1 FSR for each added unit above the single-family maximum of 0.7, 0.5, and 0.4 in zones R2.5, R5, and R7, respectively.

This upzoning provides variation in the mix of addition and division benefits along multiple dimensions. The density allowances were more valuable near the downtown core, where existing size limits were more tightly binding. Also, different zones were affected with different timing.

Vancouver, BC Canada

In Vancouver, increases in allowed densities that we will study on single-family lots occurred in two rounds. The first was the September 2018 approval of duplexes, with up to two additional rental basement suites (one per unit) permitted on nearly all single-family lots. Because duplexes under this reform could not include laneway homes (ADUs), the reform was essentially a division-only reform, in some cases with *subtraction* of square footage.

Following the Province of British Columbia's elimination of exclusive single-family zoning in November 2023, the City of Vancouver extended multiplex permissions to all residential zones. At the same time, the maximum allowed size for ADU units was increased from 0.16 to 0.25 FSR. The number of permitted units varies between three and six (and up to eight for purpose-built rental), depending on lot area and frontage. A standard 33' × 122' lot would allow three units. To allow four units, lot area would have to exceed 4,984 sq ft or frontage would have to exceed 43.6'.

With the 2023 changes, maximum allowed FSR depended on several factors. Single-family homes and duplexes were permitted maximum FSRs of 0.6 and 0.70, respectively, with an additional 0.25 FSR allowed for an ADU on single-family lots. For multiplexes (three or more units), the default maximum was 0.7, with 1.0 FSR attainable if one of the following applied: (i) all units are secured-rental housing; (ii) one unit is designated as "below-market" homeownership; or (iii) the developer pays a density bonus, which can vary from $3.00 to $140.00 per square foot of additional density depending on lot size and location. For lots of sufficient size, 7–8 rental units are also possible at an FSR of 1.0.

For Vancouver, a natural prediction is that duplex rezoning, which until late 2023 was division-only should have negative sorting. Multiplex zoning features both addition and division effects, so the prediction is more ambiguous. Laneway units, which cannot be sold, represent a particular form of addition.

3 Methodology

We wish to ask: can the spatial correlation between infill takeup and residential value per square foot within areas sharing the same city-wide upzoning be explained by the spatial distribution of the elasticity of willingness to pay with respect to lot or structure size?

A more refined version of the same question is: can the spatial correlation between infill takeup and residential value per square foot be explained by a parsimonious model of the profitability of multifamily infill versus single-family development that builds on estimates of the elasticity of willingness to pay with respect to lot or structure size?

Our approach will be to estimate at the lot level a multinomial or binomial model of the choice of which type of home to build on lots in upzoned areas. We can evaluate both the causal effect of upzoning on the choice to redevelop or not, and the relationship between the conditional choice of which type of product to build (e.g. single family with no ADU, single family with ADU, duplex, multiplex, etc.) and the expected profit of the redevelopment choice. We propose to model the profitability of different sized units based on pricing just before upzoning, both by studying relative price per square foot of single family versus grandfathered multifamily homes in affected neighbourhoods, and by studying the elasticity of price per square foot in unit size. We have shown already for the three study cities that this elasticity is highly positively correlated with neighbourhood prices and amenity levels. That is, willingness to pay for incremental square footage is highly correlated with average price levels.

Our estimation will proceed in three steps:

1. Estimate (with Attom data in the US and BC Assessment data in Vancouver) average price per square foot and the elasticity of that price with respect to unit size in the period before upzoning that allows infill housing. These values can be imputed to permitted lots and candidate lots for redevelopment using spatially weighted regression.

2. Use the elasticities and construction cost (from RS Means in the US and Butterfield in Canada) to estimate the expected profitability of different types of infill housing (e.g. duplexes, triplexes, etc.) in different neighbourhoods in each city. This profitability will be explicitly linked to the extent to which a rezoning is division-only or mixed addition and division. For example, the sale price of a half duplex on a 4,000 square foot with an allowable floor area ratio (FAR) of .8 in a neighborhood with a lower current FAR might be estimated as the pre-upzoning fitted sale price of a 1,600 square foot home on a 2,000 square foot lot. That estimate would be out-of sample, but could be fitted by estimating price elasticities with respect to lot and structure square feet among new homes. We might also adjust for estimated pre-upzoning premiums for single versus multifamily homes in the same neighborhood. The sale price of the competing single family use could likely be estimated in-sample.

3. Estimate the relationship between expected profitability both based on the second step above, and on broad characterization of a neighbourhood's upzoning as division-only or inclusive of both, and (a) the discrete choice of whether to redevelop an existing single family home and (b) the observed choice of infill housing type conditional on redevelopment, using multinomial or binomial models at the lot level, or block-or neighbourhood OLS regressions. Here we are asking the critical question of whether the correlation between price level and infill choice type *within* reform boundaries depends on the addition-division mix of reforms.

3.1 Threats to validity and robustness

Naturally upzoning boundaries might follow demand. Vancouver is a particularly useful laboratory because of the citywide nature of reform. However, we mostly study the correlation between infill choices and the addition-division mix of reforms within zones. That is, in Portland and Minneapolis, allowable uses and densities depend on location, but we are interested in the allocation of permits between newly allowed multifamily and single-family homes within zones, so the spatial endogeneity concern disappears.

Another concern is out-of-sample performance of elasticity estimates. That is, the elasticity of price per square foot and size among single family homes between 2,500 and 3,500 square feet may not provide reliable estimates of 1,250 square foot multiplex units. For that reason, we will also consider where feasible sale prices of grandfathered and nearby multifamily units in the pre-reform period to guide functional form choices for extrapolation.

3.2 Feasibility

Numerous jurisdictions, including those enumerated above as well as Seattle and now all of California have been upzoned and permits have been issued allowing infill housing for several years. With price and permitting data and estimates of construction costs by type (e.g. available from RS Means in the US and Butterfield Property Advisers in Vancouver), as well as neighbourhood demographic and standard geographic amenity measures, we have the necessary data to implement the above estimation strategy. We have preliminary results already at a coarse neighbourhood level for the three focal cities, each of which provides publicly available (if messy) data on permits.

3.3 Study Timeline

- Months 1-4: Refine the economic model and estimation strategy and clean construction cost, price, and permitting data for additional US cities
- Months 5-8: Estimate the model as described above.
- Months 9-12: Document results and write both an academic paper and a policy brief per Arnold's preferences.
- Months 12-15: Revise and submit paper to an academic journal and disseminate policy brief to policymakers and the public.
- Months 1-18: Present the paper to academic and practitioner audiences.

4 Budget

Our initial request is for \$150,000. The primary cost drivers are data acquisition and personnel. We will use funding to support data acquisition and a graduate research assistant. We have already acquired Attom data, but expanding the time covered would be a natural and large use of funds. We also would like to acquire detailed RS Means data. We will also fund a graduate research assistant as a cost of roughly \$30,000 for one year.

5 Team

The team is the three authors plus an RA, likely but not necessarily our doctoral student Daryl Larsen. Both Davidoff and Somerville have both extensive publication records and have participated in public discussions and worked with government on zoning reform. We were part of the team that the BC government tapped to model multiplex reform (Bergmann et al. (2023)). Helsley has published extensively on the economics of land use regulation and has served as Dean of the Sauder School of Business, and as a faculty chair of planning for a major University expansion in Surrey, B.C.

References

Bergmann, Jens von et al. (Dec. 2023). *SSMUH and TOA Scenarios in British Columbia.* Technical Report. Ministry of Housing, Province of British Columbia. URL: `https://www2.gov.bc.ca/assets/gov/housing-and-tenancy/tools-for-government/local-governments-and-housing/ssmuh_toa_scenarios.pdf`.

Brueckner, Jan K. and Sarah Thomaz (2024). “ADUs in Los Angeles: Where are they located and by how much do they raise property value?” In: *Real Estate Economics* 52.3, pp. 885–907. DOI: https://doi.org/10.1111/1540-6229.12471. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/1540-6229.12471.

Dong, Hongwei and J. Andy Hansz (2019). “Zoning, density, and rising housing prices: A case study in Portland, Oregon”. In: *Urban Studies* 56.16, pp. 3486–3503.

Marantz, Nicholas J., Christopher S. Elmendorf, and Youjin B. Kim (Jan. 2023). *Evaluating California's Accessory Dwelling Unit Reforms: Preliminary Evidence and Lessons for State Governments.* Research Report. NYU Furman Center. URL: https://furmancenter.org/files/Evaluating_California%E2%80%99s_Accessory_Dwelling_Unit_Reforms_508.pdf.